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Effects of implementing Pressure Ulcer Prevention Practice Guidelines (PUPPG) in the prevention of pressure ulcers among hospitalized elderly patients: A systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-043042
Article Type:	Protocol
Date Submitted by the Author:	23-Nov-2020
Complete List of Authors:	Wung Buh, Amos; University of Ottawa, Interdisciplinary School of Health Sciences Mahmoud, Hassan; University of Ottawa, School of Epidemiology and Public Health, Faculty of Medicine Chen, Wenjun; University of Ottawa, School of Nursing; Central South University, Xiangya School of Nursing McInnes, Matthew D. F.; University of Ottawa, School of Epidemiology and Public Health, Faculty of Medicine Fergusson, Dean; Ottawa Hospital Research Institute, Medicine
Keywords:	Geriatric dermatology < DERMATOLOGY, GERIATRIC MEDICINE, Protocols & guidelines < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, ORTHOPAEDIC & TRAUMA SURGERY, WOUND MANAGEMENT

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TITLE PAGE

Title: Effects of implementing Pressure Ulcer Prevention Practice Guidelines (PUPPG) in the prevention of pressure ulcers among hospitalized elderly patients: A systematic review protocol

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Word count - Abstract: 285; Text: 2356

Number of figures and tables: 0

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Abstract

Introduction

Pressure ulcers are serious and potentially life-threatening problems across all age groups and across all medical specialties and care settings. The hospitalized elderly population is the most common group to develop pressure ulcers. This study aims to systematically review studies implementing pressure ulcer prevention strategies recommended in the Pressure Ulcer Prevention Practice Guidelines for the prevention of pressure ulcers among hospitalized elderly patients globally.

Methods and analysis

A systematic review of all studies that have assessed the use of pressure ulcer prevention strategies in hospital settings among hospitalized elderly patients shall be conducted. A comprehensive search of all published articles in Medline Ovid, CINAHL, PubMed, Embase, Cochrane library, Scopus and Web of Science will be done using terms such as pressure ulcers, prevention strategies, elderly patients, and hospital. Studies will be screened for eligibility through title, abstract and full text by two independent reviewers. Study quality and risk of bias will be assessed using the Joanna Briggs Institute for Meta-Analysis of Statistics Assessment and Review Instrument. If sufficient data is available, a meta–analysis will be conducted to synthesize the effect size reported as odds ratio with 95% CI using both fixed and random effect models. I^2 statistics and visual inspection of the forest plots will be used to assess heterogeneity and identify the potential sources of heterogeneity. Publication bias will be assessed by visual inspections of funnel plots and Egger's test.

Ethics and dissemination

No formal ethical approval or consent is required as no primary data will be collected. We aim to publish the research findings in a peer–reviewed scientific journal to promote knowledge transfer, as well as in conferences, seminars, congresses or symposia in a traditional manner.

PROSPERO registration number: CRD42019129088.

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Strengths and limitations of this study

- This is a systematic review and metal analysis of randomized controlled trials.
- This review will be the first to synthesize the evidence regarding the effectiveness of guidelines used in pressure ulcer prevention for elderly patients in hospitals and offer the highest level of evidence for informed decisions on use of PUPPG in prevention pressure ulcers in the elderly patients in hospital.
- There may be heterogeneity of interventions used on eligible studies and incomplete information reported about the interventions in the literature which could limit our ability to statistically compare the effectiveness of interventions.
- The main limitation of this review might be scarcity of randomized controlled trials on the use of PUPPG for preventing pressure ulcers in elderly patients, publication bias and methodological quality of grey literature that shall be found.

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Background

Pressure ulcers (PU) also known as pressure injuries are areas of localized damage to the skin and/or underlying structures due to pressure and/or friction and shear[1]. They are serious and potentially life-threatening problems across all age groups from the very young to the very old and across all medical specialties and care settings[2]. It has been documented that hospital admissions due to PU are 75% higher than admissions for any other medical conditions and that, the consequences of PU development in hospitalized patients are particularly serious[2]. Patients with hospital admission PU are three times more likely to be discharged to long-term care facilities and mortality of these patients is twice that of patients without hospital admission PU[3]. The cost of treatment of PU is 2.5 times than its prevention, and PU increases the length of stay in the hospital from 4 to 30 days, decreases quality of life, and increases pain, morbidity, and mortality[4].

On international level, hospital acquired pressure ulcers (sometimes called decubitus ulcers) are very common[5]. Although many of these cases are preventable, their point prevalence in Canadian hospitals for example is measured to be 25.1%[6]. Unfortunately, the high rates of such condition are associated with subsequent high burden on the healthcare system and the national economy considering the high cost of their management, and the frequent occurrence of associated significant morbidity and mortality[5]. According to the Ontario Case Costing Initiative database in 2013 using the European Pressure Ulcer Advisory Panel (EPUAP) staging system, it was estimated that the cost of management of stage II ulcer is up to 40 thousand dollars and can reach more than double this price for managing a single case of stage IV ulcer[7]. A good example of the burden that pressure ulcer add to the national economy was measured in United states; it was estimated that hospital acquired pressure ulcers increase the financial expenses on health care systems between 6 and 15 billion dollars annually[8].

The National Pressure Ulcer Advisory Panel (NPUAP), the European Pressure Ulcer Advisory Panel (EPUAP) and the Pan Pacific Pressure Injury Alliance (PPPIA)[9] have defined pressure ulcer as a ‘lesion or a trauma to the skin and/or underlying tissue usually over a bony prominence and is the result of undiminished pressure, or pressure combination with shear, friction and moisture’. It is a degenerative progress attributable to biological tissues (skin and underlying tissues) being exposed to pressure and shearing forces. The pressure constrains the proper blood circulation and causes cell death, tissue necrosis and the development of ulcers[9]. While the quality of pressure ulcer prevention and treatment has increased considerably over the past years, pressure

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ulcers remains a global concern because of its frequency of occurrence and negative consequences for patients and families as well as for the healthcare system[10]. Incidence of pressure ulcers for hospitalized patients ranges from 9% to 18%, among which the elderly population appears to be the most common group to develop the ulcers[11]. At the same time, many elderly patients are more vulnerable to be 'stuck' at a certain stage of pressure ulcer for a long period of time and sometimes for the remainder of their lives[12]. This may result in longer length of hospital stay, heavier burdens for the health care system and family members, worst quality of life for elderly patients, which may also influence their mental health such as emotional stability[13,14].

NPUAP, EPUAP and PPPIA[9] developed the Pressure Ulcer Prevention Practice Guideline (PUPPG) which involves a range of evidence-based recommendations for pressure ulcers prevention that could be applied by healthcare professionals globally. Frequently used pressure ulcer prevention strategies recommended in this guideline includes pressure ulcer risk assessment, regular repositioning, prevention management plan, appropriate use of support surfaces and protection, continence management, patient education, skin protection, nutritional assessment, and adequate nutrition[15]. It also includes some recommendations specifically for elderly people - 'protect aged skin from skin injury associated with pressure and shear forces', taking into consideration that an aged person's skin is vulnerable[15].

A number of studies have been conducted on the implementation of pressure ulcer prevention strategies among hospitalized patients. One cluster randomized trial conducted in Canada revealed that multi-disciplinary pressure ulcer prevention groups are more cost effective than usual care and yields no significant improvement in the treatment of pressure ulcers [16]. Despite the existence of the guidelines on the prevention of pressure ulcers, their effective utilization in preventing pressure ulcers among hospitalized elderly patients varies in settings and countries. Also, although a number of studies have assessed strategies used in preventing pressure ulcers, there appears to be little or no information on systematic reviews that have assessed the effectiveness of guidelines used in pressure ulcer prevention for elderly patients in hospitals. This study therefore, aims to systematically review studies implementing pressure ulcer prevention strategies recommended in the PUPPG for the prevention of pressure ulcers among hospitalized elderly patients globally.

Objective:

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The objective of this review is to assess the effectiveness of each of the strategies included in the PUPPG guideline in reducing the incidence and prevalence of hospital acquired pressure ulcers in hospitalized elderly patients in comparison to no strategy (usual practice), or other strategies. The review question is: what is the effectiveness of implementing each of the pressure ulcer prevention strategies included in the Pressure Ulcer Prevention Practice Guideline (PUPPG) in decreasing the incidence and prevalence of pressure ulcers among hospitalized elderly patients compared to no strategies (basic usual care) or different prevention strategies?

Methods

Study design:

This will be a systematic review and meta-analysis of published and unpublished studies that have assessed the use of pressure ulcer prevention strategies in hospital settings among hospitalized elderly patients. The systematic review protocol has been developed and reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) criteria[17].

Inclusion criteria

Population included:

This systematic review will focus on studies that involved all vitally stable (not admitted in the Intensive Care unit (ICU)) bed ridden hospitalized patients aged 60 or above.

Interventions:

All studies that assessed the effect of Pressure Ulcers preventive strategies found in the PUPPG, that were implemented on vitally stable bed ridden hospitalized patients aged 60 and above with an aim to decrease the occurrence of pressure ulcers, will be included in this review. Interventions will be limited to use of risk assessment, skin assessment, skin care, nutrition, position and repositioning, education and training, and medical devices care.

Comparator:

Interventions will be compared with other strategies to identify the most effective among them and/or will also be compared to no interventions (regular basic management).

Outcomes:

In this study the primary outcome will be directly related to the incidence of the disease among elderly hospitalized patients (incidence shall be considered as the proportion of hospitalized

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patients who developed pressure ulcers while in hospital). Included studies must measure study duration related incidence of the disease and /or its point prevalence and /or stage of pressure ulcer(severity) as a measure of the effectiveness of the preventive strategies.

Types of studies:

We will focus only on Quantitative studies - experimental and quasi-experimental studies. These might include randomized and non-randomized controlled trials in addition to comparative and before and after studies.

Language:

Only studies written in English will be included in this systematic review.

Search strategy

We will use a three-step strategy to find published and unpublished studies on pressure ulcers and their management. Firstly, we will conduct an initial search through the Medline Ovid database using an analysis of text words found in the title and abstract, and the index terms used to describe the article. Secondly, we will use identified keywords and index terms to search for studies in identified databases. Finally, we will use the reference list of selected studies from the first and second searches to look for additional studies not found in the databases. For this study, we will consider only studies either published or unpublished in English.

The databases that shall be searched for this review will include Medline Ovid, CINAHL, PubMed, Embase, Cochrane library, Scopus and Web of Science. See Appendix 1 for the example searching strategy and results in Medline (Ovid). All these databases will provide published studies. To find unpublished studies on our topic, we will use Google, Grey Literature reports and the Centers for Disease Control and Prevention.

The keywords we will use for our initial searches in Medline Ovid will include “pressure ulcers”, “pressure sore”, “bed sore”, “pressure injuries”, “prevention strategies”, “elderly patients”, and “hospital”.

Study screening and selection

The titles, abstracts and full-text of studies selected for this study will be reviewed by two independent researchers to identify studies that potentially meet the inclusion criteria outlined above. The Covidence software will be used for title, abstract and full-text screening. After importing references and inclusion/exclusion criteria into the Covidence software, two independent reviewers will screen titles of included studies according to the eligible criteria.

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Conflicts between those two reviewers will be resolved through discussion with a third reviewer. The same procedures shall be utilized for abstract screening. Following the abstract screening, full texts of these potentially eligible studies will be retrieved and independently assessed for eligibility by two reviewers. Any disagreement between the two reviewers over the eligibility of a particular study will also be resolved through discussion with the third reviewer. The process of study selection will be reported using the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta–Analyses) flow diagram[17].

Assessment of methodological quality

Two independent reviewers will be used to assess the methodological validity of the quantitative papers that will be selected for retrieval prior to their inclusion in the review using standard critical appraisal tools from the Joanna Briggs Institute for Meta-Analysis of Statistics Assessment and Review Instrument (See Appendix 2). All disagreement between the two reviewers shall be settled through discussions.

Data Extraction:

After screening and selecting studies, key information from those studies will be extracted into an excel sheet for further analysis. We shall use a data extraction tool adapted from the standardized data extraction tool from the Joanna Briggs Institute Meta-Analysis of Statistics Assessment and Review Instrument (JBI-MAStARI). Considering the information we will need for the data synthesis of our study, we shall use the JBI-MAStARI to develop a data extraction tool specifically for quantitative research data extraction (see Appendix 3). The tool will be used to extract: 1) Study characteristics of reviewed papers, such as authors, year of publication, journal; 2) Methods of the study, including study design (RCT, quasi-RCT, Longitudinal, retrospective), research purpose and/or questions; 3) Participant characteristics, country where the study took place, setting, population, sample size, age, sex, ethnicity, socioeconomic status and/or education level; 4) Pressure ulcer prevention strategies utilized in experimental group and control group (if applicable), 5) Outcome measures and results and 6) Conclusions of reviewed papers and any comments from reviewers. Two reviewers will independently perform data extraction. Authors of reviewed papers will be contacted in case of any missing details about their studies.

Data synthesis

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A meta-analysis of outcomes combining various studies included in the review shall be done. We will assess statistical heterogeneity with I^2, which will indicate the percentage of the total variation across studies: 0% - 40% low heterogeneity, 30% - 60% moderate heterogeneity, 50% - 90% may represent substantial heterogeneity, and 75% - 100% is considerable heterogeneity. If there is a substantial amount of heterogeneity (75%), then sources of heterogeneity will be examined through subgroup and sensitivity analyses. We will also use Chi-square test to test the heterogeneity and consider P-values < 0.05 as statistically significant. A fixed-effects model will be selected for significant homogeneous studies; otherwise we will apply a random-effects model. All outcomes will be summarized using odds ratios (OR) and 95% confidence intervals (CI). An OR<1 will represent a lower rate of outcome among the group of patients who were treated following the guidelines. Publication bias will be assessed by visual inspections of funnel plots and Egger's test.

We will also provide a narrative synthesis of the findings from the included studies. The narrative synthesis shall be structured by describing the studies according to the type of intervention used. This will include the three categories of interventions recommend in the PUPPG guideline [9]:

1. Prevention of pressure ulcers, including risk factors and risk assessment, skin and tissue assessment, preventive skin care and emerging therapies for prevention of pressure ulcers;

2. Interventions for prevention and treatment of pressure ulcers, such as nutrition in pressure ulcer prevention and treatment, repositioning and early mobilization, repositioning to prevent and treat pressure ulcers, support surface and medical device related pressure ulcers;

3. Treatment of pressure ulcers, e.g. assessment of pressure ulcers and monitoring of healing, pain assessment and treatment, wound care, assessment and treatment of infection and biofilms, wound dressings for treatment of pressure ulcers and surgery for pressure ulcers. Results will be presented in tables, figures and graphs followed by discussion.

Ethics and Dissemination:

This review will only use published literature and will not recruit participants. Therefore, no formal ethical approval or consent is necessary. It is anticipated that this systematic review will provide a detailed summary of the evidence of the effectiveness of the PUPPG in preventing the occurrence of pressure ulcers among elderly patients in hospital. It is also expected that the study

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will provide recommendations on the best pressure ulcer preventive strategies applicable in health care settings. We aim to publish the research findings in a peer–reviewed scientific journal to promote knowledge transfer, as well as in various media, such as: conferences, seminars, congresses or symposia in a traditional manner.

LIST OF ABBREVIATIONS

NPUAP: National Pressure Ulcer Advisory Panel

EPUAP: European Pressure Ulcer Advisory Panel

PPPIA: Pan Pacific Pressure Injury Alliance

PUPPG: Pressure Ulcer Prevention Practice Guidelines

PRISMA-P: Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols

PROSPERO: International Prospective Register of Systematic Reviews

JBI: Joanna Briggs Institute

JBI-MASTARI: Joanna Briggs Institute Meta-Analysis of Statistics Assessment and Review instrument

GRADE: Grading of Recommendations Assessment, Development and Evaluation

DECLARATIONS

Publication Bias and Small Study Effects

Publication bias will be assessed in all analyses synthesizing 10 or more studies to ensure adequate power in the analysis[18]. For investigation of the effect of small studies and publication bias, data from included studies will be entered into a funnel plot asymmetry test if we have at least 10 studies in the meta–analysis. Egger's statistical test will be implemented using STATA/SE version 13 (Stata Corp, College Station, TX).

Confidence in Cumulative Evidence

The quality of supporting evidence will be assessed by the Grades of Recommendation, Assessment, Development and Evaluation (GRADE)[19].

Amendments

This is an original research protocol, as opposed to an amendment of a previously completed protocol. Any modifications made to this protocol will be documented and updated via PROSPERO and stated in the final review manuscript.

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Consent for publication: Consent to publish is not needed as data that shall be used in this review shall be available in the public domain. Also, we will not use any details, images or videos related to individual participants in the review.

Availability of data and materials: Not applicable

Competing interests: The authors declare that they have no competing interests

Funding: This work was supported by Hunan Provincial Key Laboratory of Nursing, grant number (2017TP1004), Hunan Provincial Science and Technology Department, China.

Author Contributions

AB, HM and WC contributed to the conception of the research question. HM, AB, WC, MM and DF contributed to the development of search strategies, eligibility criteria and methodology for data synthesis. HM, AB, WC, MM and DF contributed to drafting of the protocol and provided approval for the final version of this protocol. HM, AB and WC will work in duplicate to screen the titles and abstracts of all the materials obtained using the search strategy to exclude the articles that do not meet the eligibility criteria. HM, AB and WC will evaluate the potentially eligible studies with the full text and further exclude studies with documentation of the reason for exclusion. All authors will contribute to the bias assessment strategy and data extraction criteria. HM, AB and WC will independently extract data from the included studies. HM, AB and WC will analyze the data and draft the manuscript. All authors will read, provide feedback and approve the final manuscript.

Acknowledgements: The authors would like to thank Lindsey Sikora (librarian) for counseling in developing the searching strategies.

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Appendices

Appendix 1

Searching Strategies and Results in Medline (Ovid)

Database: Ovid MEDLINE(R) and Epub Ahead of Print, In Process & Other Non-Indexed Citations and Daily <1946 to March 12, 2019>
Search Strategy:

---

1 Pressure Ulcer/ (11656)
2 (pressure adj3 sore*).ti,ab. (3032)
3 (pressure adj3 ulcer*).ti,ab. (7268)
4 (bed adj3 sore*).ti,ab. (221)
5 (decubitus adj3 ulcer*).ti,ab. (1787)
6 (pressure adj3 injur*).ti,ab. (2048)
7 or/1-6 (17175)
8 aged/ or "aged, 80 and over"/ (2913834)
9 elder*.ti,ab. (237584)
10 aged.ti,ab. (516384)
11 geriatric*.ti,ab. (43081)
12 or/8-11 (3320603)
13 Practice Guideline/ or Guideline/ or Guideline Adherence/ (60850)
14 guideline*.ti,ab. (306303)
15 pathway*.ti,ab. (997675)
16 protocol*.ti,ab. (390728)
17 PUPPG.ti,ab. (0)
18 recommendation*.ti,ab. (227666)
19 or/13-18 (1857233)
20 7 and 12 and 19 (544)

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Appendix 2

Joanna Briggs Institute for Meta-Analysis of Statistics Assessment and Review Instruments

1. JBI critical appraisal checklist for randomized control trials

Reviewer…….. Date……..

Author…….. Year……..

		Yes	No	Unclear	Not applicable
1	Was the assignment and treatment groups truly random?				
2	Were participants blind to treatment allocation?				
3	Was the allocation to treatment groups concealed from the allocator?				
4	Were the outcomes of people who withdrew described and included in the analysis?				
5	Were those assessing outcomes blind to the treatment allocation?				
6	Were the control and treatment groups comparable at entry?				
7	Were groups treated identically other than for the named interventions?				
8	Were outcomes measured in the same way for all groups?				
9	Were outcomes measured in a reliable way?				
10	Was appropriate statistical analysis used?				

Overall appraisal: include…… exclude……. Seek further info…….

Comments (include reasons for exclusion)……..

2. JBI critical appraisal checklist for non-randomized control trials

Reviewer…….. Date……..

Author…….. Year……..

1	Was study based on random or pseudo sample?				

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2	Were the criteria for inclusion in the sample clearly defined?				
3	Were confounding factors identified and strategies to deal with them stated?				
4	Were outcomes assessed using objective criteria?				
5	If comparisons are being made, was there sufficient description of the groups?				
6	Was follow up carried out over a sufficient time period?				
7	Were the outcomes of people who withdrew described and included in the analysis?				
8	Were outcomes measured in a reliable way?				
9	Was appropriate statistical analysis used?				

Overall appraisal: include…… exclude……. Seek further info…….

Comments (include reasons for exclusion)……..

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Appendix 3

Data Extraction Form for Quantitative Research

Study Characteristics

1. **Author:**
2. **Year of publication:**
3. **Journal:**
4. **Record number:**

Study Method

1. **Study Design:**

RCT	Quasi-RCT	Longitudinal	Retrospective	Observational	Other

2. **Purpose:**
3. **Research Questions:**

Participant Characteristics

Country where the study took place	Setting	Population	Sex	Age	Sample size	Ethnicity (if applicable)	socioeconomic status (if applicable)	education level (if applicable)

Pressure Ulcer Prevention Strategies

Strategy 1:

Strategy 2:

Strategy 3:

Outcome Measures

Outcome Description	Scale/measure

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Study results

Dichotomous data

Outcome	Intervention ( ) number / total number	Intervention ( )number / total number

Continuous data

Outcome	Intervention ( ) mean & SD (number)	Intervention ( ) mean & SD (number)

Authors' conclusions:

Comments:

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PRISMA-P 2015 Checklist

This checklist has been adapted for use with protocol submissions to *Systematic Reviews* from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☑	☐	Page2/21, line 8
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☐	
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☑	☐	Page 4/21, line 47
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☑	☐	Page 2/21, lines 10-55
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☑	☐	Page 13/21, lines 3-21
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☑	☐	Page 12/21, lines 36-41
Support					
Sources	5a	Indicate sources of financial or other support for the review	☑	☐	Page 12/21, lines 51-53
Sponsor	5b	Provide name for the review funder and/or sponsor	☑	☐	Page 12/21, lines 53
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☑	☐	Page 12/21, lines 51-53
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	☑	☐	Page 6-7/21
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☑	☐	Page 8/21, lines 3-14

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☑	☐	Page 8/21, lines 20-55; Page 9/21, lines 3-18
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☑	☐	Page 9/21, lines 32-40
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☑	☐	Page 9/21, lines 20-31
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☑	☐	Page 9/21, line 52
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☑	☐	Page 9/21, lines 48-55; page 10/21, lines 3-14
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☑	☐	Page 10/21, lines 26-50
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☑	☐	Page 8/21, lines 31-50
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☑	☐	Page 8/21, lines 52-55; page 9/21, lines 3-7
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☑	☐	Page 10/21, lines- 15-24
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☑	☐	Page 11/21, lines 3-43
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☑	☐	Page 11/21, lines 3-14

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☑	☐	Page 11-21,lines 14-21
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☑	☐	Page 11/21, lines 22-43
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	☑	☐	Page 12/21, lines 32-41
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	☑	☐	Page 12/21, lines 43-47

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BMJ Open

Effects of implementing Pressure Ulcer Prevention Practice Guidelines (PUPPG) in the prevention of pressure ulcers among hospitalized elderly patients: A systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-043042.R1
Article Type:	Protocol
Date Submitted by the Author:	08-Feb-2021
Complete List of Authors:	Wung Buh, Amos; University of Ottawa, Interdisciplinary School of Health Sciences Mahmoud, Hassan; University of Ottawa, School of Epidemiology and Public Health, Faculty of Medicine Chen, Wenjun; University of Ottawa, School of Nursing; Central South University, Xiangya School of Nursing McInnes, Matthew D. F.; University of Ottawa, School of Epidemiology and Public Health, Faculty of Medicine; Ottawa Hospital Research Institute Clinical Epidemiology Program Fergusson, Dean; Ottawa Hospital Research Institute Clinical Epidemiology Program, Medicine
Primary Subject Heading:	Geriatric medicine
Secondary Subject Heading:	Geriatric medicine, Evidence based practice, Nursing
Keywords:	Geriatric dermatology < DERMATOLOGY, GERIATRIC MEDICINE, Protocols & guidelines < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, ORTHOPAEDIC & TRAUMA SURGERY, WOUND MANAGEMENT

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TITLE PAGE

Title: Effects of implementing Pressure Ulcer Prevention Practice Guidelines (PUPPG) in the prevention of pressure ulcers among hospitalized elderly patients: A systematic review protocol

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Word count - Abstract: 296; Text: 2348

Number of figures and tables: 0

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Abstract

Introduction

Pressure ulcers are serious and potentially life-threatening problems across all age groups and across all medical specialties and care settings. The hospitalized elderly population is the most common group to develop pressure ulcers. This study aims to systematically review studies implementing pressure ulcer prevention strategies recommended in the Pressure Ulcer Prevention Practice Guidelines (PUPPG) for the prevention of pressure ulcers among hospitalized elderly patients globally.

Methods and analysis

A systematic review of all studies that have assessed the use of pressure ulcer prevention strategies in hospital settings among hospitalized elderly patients shall be conducted. A comprehensive search of all published articles in Medline Ovid, Cumulative Index to Nursing and Allied Health Literature (CINAHL), PubMed, Embase, Cochrane library, Scopus and Web of Science will be done using terms such as pressure ulcers, prevention strategies, elderly patients, and hospital. Studies will be screened for eligibility through title, abstract and full text by two independent reviewers. Study quality and risk of bias will be assessed using the Joanna Briggs Institute for Meta-Analysis of Statistics Assessment and Review Instrument. If sufficient data is available, a meta–analysis will be conducted to synthesize the effect size reported as odds ratio with 95% confidence intervals (CI) using both fixed and random effect models. I^2 statistics and visual inspection of the forest plots will be used to assess heterogeneity and identify the potential sources of heterogeneity. Publication bias will be assessed by visual inspections of funnel plots and Egger's test.

Ethics and dissemination

No formal ethical approval or consent is required as no primary data will be collected. We aim to publish the research findings in a peer–reviewed scientific journal to promote knowledge transfer, as well as in conferences, seminars, congresses or symposia in a traditional manner.

PROSPERO registration number: CRD42019129088.

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Strengths and limitations of this study

- This is a systematic review and meta-analysis of randomized controlled trials.
- This review will be the first to synthesize the evidence regarding the effectiveness of guidelines used in pressure ulcer prevention for elderly patients in hospitals and offer the highest level of evidence for informed decisions on use of PUPPG in prevention pressure ulcers in the elderly patients in hospital.
- There may be heterogeneity of interventions used on eligible studies and incomplete information reported about the interventions in the literature which could limit our ability to statistically compare the effectiveness of interventions.
- The main limitation of this review might be scarcity of randomized controlled trials on the use of PUPPG for preventing pressure ulcers in elderly patients, publication bias and methodological quality of grey literature that shall be found.

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Background

Pressure ulcers (PU) also known as pressure injuries are areas of localized damage to the skin and/or underlying structures due to pressure and/or friction and shear[1]. They are serious and potentially life-threatening problems across all age groups from the very young to the very old and across all medical specialties and care settings[2]. It has been documented that hospital admissions due to PU are 75% higher than admissions for any other medical conditions and that, the consequences of PU development in hospitalized patients are particularly serious[2]. Patients with hospital admission PU are three times more likely to be discharged to long-term care facilities and mortality of these patients is twice that of patients without hospital admission PU[3]. The cost of treatment of PU is 2.5 times than its prevention, and PU increases the length of stay in the hospital from 4 to 30 days, decreases quality of life, and increases pain, morbidity, and mortality[4].

On international level, hospital acquired pressure ulcers (sometimes called decubitus ulcers) are very common[5]. Although many of these cases are preventable, their point prevalence in Canadian hospitals for example is measured to be 25.1%[6]. Unfortunately, the high rates of such condition are associated with subsequent high burden on the healthcare system and the national economy considering the high cost of their management, and the frequent occurrence of associated significant morbidity and mortality[5]. According to the Ontario Case Costing Initiative database in 2013 using the European Pressure Ulcer Advisory Panel (EPUAP) staging system, it was estimated that the cost of management of stage II ulcer is up to 40 thousand dollars and can reach more than double this price for managing a single case of stage IV ulcer[7]. A good example of the burden that pressure ulcer add to the national economy was measured in United states; it was estimated that hospital acquired pressure ulcers increase the financial expenses on health care systems between 6 and 15 billion dollars annually[8].

The National Pressure Ulcer Advisory Panel (NPUAP), the European Pressure Ulcer Advisory Panel (EPUAP) and the Pan Pacific Pressure Injury Alliance (PPPIA)[9] have defined pressure ulcer as a ‘lesion or a trauma to the skin and/or underlying tissue usually over a bony prominence and is the result of undiminished pressure, or pressure combination with shear, friction and moisture’. It is a degenerative progress attributable to biological tissues (skin and underlying tissues) being exposed to pressure and shearing forces. The pressure constrains the proper blood circulation and causes cell death, tissue necrosis and the development of ulcers[9]. While the quality of pressure ulcer prevention and treatment has increased considerably over the past years, pressure

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ulcers remains a global concern because of its frequency of occurrence and negative consequences for patients and families as well as for the healthcare system[10]. Incidence of pressure ulcers for hospitalized patients ranges from 9% to 18%, among which the elderly population appears to be the most common group to develop the ulcers[11]. At the same time, many elderly patients are more vulnerable to be 'stuck' at a certain stage of pressure ulcer for a long period of time and sometimes for the remainder of their lives[12]. This may result in longer length of hospital stay, heavier burdens for the health care system and family members, worst quality of life for elderly patients, which may also influence their mental health such as emotional stability[13,14].

NPUAP, EPUAP and PPPIA[9] developed the Pressure Ulcer Prevention Practice Guideline (PUPPG) which involves a range of evidence-based recommendations for pressure ulcers prevention that could be applied by healthcare professionals globally. Frequently used pressure ulcer prevention strategies recommended in this guideline includes pressure ulcer risk assessment, regular repositioning, prevention management plan, appropriate use of support surfaces and protection, continence management, patient education, skin protection, nutritional assessment, and adequate nutrition[15]. It also includes some recommendations specifically for elderly people - 'protect aged skin from skin injury associated with pressure and shear forces', taking into consideration that an aged person's skin is vulnerable[15].

A number of studies have been conducted on the implementation of pressure ulcer prevention strategies among hospitalized patients. One cluster randomized trial conducted in Canada revealed that multi-disciplinary pressure ulcer prevention groups are more cost effective than usual care and yields no significant improvement in the treatment of pressure ulcers [16]. Despite the existence of the guidelines on the prevention of pressure ulcers, their effective utilization in preventing pressure ulcers among hospitalized elderly patients varies in settings and countries. Also, although a number of studies have assessed strategies used in preventing pressure ulcers, there appears to be little or no information on systematic reviews that have assessed the effectiveness of guidelines used in pressure ulcer prevention for elderly patients in hospitals. This study therefore, aims to systematically review studies implementing pressure ulcer prevention strategies recommended in the PUPPG for the prevention of pressure ulcers among hospitalized elderly patients globally.

Objective:

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The objective of this review is to assess the effectiveness of each of the strategies included in the PUPPG guideline in reducing the incidence and prevalence of hospital acquired pressure ulcers in hospitalized elderly patients in comparison to no strategy (usual practice), or other strategies. The review question is: what is the effectiveness of implementing each of the pressure ulcer prevention strategies included in the Pressure Ulcer Prevention Practice Guideline (PUPPG) in decreasing the incidence and prevalence of pressure ulcers among hospitalized elderly patients compared to no strategies (basic usual care) or different prevention strategies?

Methods

Study design:

This will be a systematic review and meta-analysis of published and unpublished studies that have assessed the use of pressure ulcer prevention strategies in hospital settings among hospitalized elderly patients. The systematic review protocol has been developed and reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols (PRISMA-P) criteria (see Appendix 1)[17].

Inclusion criteria

Population included:

This systematic review will focus on studies that involved all vitally stable (not admitted in the Intensive Care unit (ICU)) bed ridden hospitalized patients aged 60 or above.

Interventions:

All studies that assessed the effect of Pressure Ulcers preventive strategies found in the PUPPG, that were implemented on vitally stable bed ridden hospitalized patients aged 60 and above with an aim to decrease the occurrence of pressure ulcers, will be included in this review. Interventions will be limited to use of risk assessment, skin assessment, skin care, nutrition, position and repositioning, education and training, and medical devices care.

Comparator:

Interventions will be compared with other strategies to identify the most effective among them and/or will also be compared to no interventions (regular basic management).

Outcomes:

In this study the primary outcome will be directly related to the incidence of the disease among elderly hospitalized patients (incidence shall be considered as the proportion of hospitalized

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patients who developed pressure ulcers while in hospital). Included studies must measure study duration related incidence of the disease and /or its point prevalence and /or stage of pressure ulcer(severity) as a measure of the effectiveness of the preventive strategies.

Types of studies:

We will focus only on Quantitative studies - experimental and quasi-experimental studies. These might include randomized and non-randomized controlled trials in addition to comparative and before and after studies.

Language:

Only studies written in English will be included in this systematic review.

Search strategy

We will use a three-step strategy to find published and unpublished studies on pressure ulcers and their management. Firstly, we will conduct an initial search through the Medline Ovid database using an analysis of text words found in the title and abstract, and the index terms used to describe the article. Secondly, we will use identified keywords and index terms to search for studies in identified databases. Finally, we will use the reference list of selected studies from the first and second searches to look for additional studies not found in the databases. For this study, we will consider only studies either published or unpublished in English.

The databases that shall be searched for this review will include Medline Ovid, CINAHL, PubMed, Embase, Cochrane library, Scopus and Web of Science. See Appendix 2 for the example searching strategy and results in Medline (Ovid). All these databases will provide published studies. To find unpublished studies on our topic, we will use Google, Grey Literature reports and the Centers for Disease Control and Prevention.

The keywords we will use for our initial searches in Medline Ovid will include “pressure ulcers”, “pressure sore”, “bed sore”, “pressure injuries”, “prevention strategies”, “elderly patients”, and “hospital”.

Study screening and selection

The titles, abstracts and full-text of studies selected for this study will be reviewed by two independent researchers to identify studies that potentially meet the inclusion criteria outlined above. The Covidence software will be used for title, abstract and full-text screening. After importing references and inclusion/exclusion criteria into the Covidence software, two independent reviewers will screen titles of included studies according to the eligible criteria.

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Conflicts between those two reviewers will be resolved through discussion with a third reviewer. The same procedures shall be utilized for abstract screening. Following the abstract screening, full texts of these potentially eligible studies will be retrieved and independently assessed for eligibility by two reviewers. Any disagreement between the two reviewers over the eligibility of a particular study will also be resolved through discussion with the third reviewer. The process of study selection will be reported using the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta–Analyses) flow diagram[17].

Assessment of methodological quality

Two independent reviewers will be used to assess the methodological validity of the quantitative papers that will be selected for retrieval prior to their inclusion in the review using standard critical appraisal tools from the Joanna Briggs Institute for Meta-Analysis of Statistics Assessment and Review Instrument (See Appendix 3). All disagreement between the two reviewers shall be settled through discussions.

Data Extraction:

After screening and selecting studies, key information from those studies will be extracted into an excel sheet for further analysis. We shall use a data extraction tool adapted from the standardized data extraction tool from the Joanna Briggs Institute Meta-Analysis of Statistics Assessment and Review Instrument (JBI-MAStARI). Considering the information we will need for the data synthesis of our study, we shall use the JBI-MAStARI to develop a data extraction tool specifically for quantitative research data extraction (see Appendix 4). The tool will be used to extract: 1) Study characteristics of reviewed papers, such as authors, year of publication, journal; 2) Methods of the study, including study design (randomized control trial (RCT), quasi-RCT, Longitudinal, retrospective), research purpose and/or questions; 3) Participant characteristics, country where the study took place, setting, population, sample size, age, sex, ethnicity, socioeconomic status and/or education level; 4) Pressure ulcer prevention strategies utilized in experimental group and control group (if applicable), 5) Outcome measures and results and 6) Conclusions of reviewed papers and any comments from reviewers. Two reviewers will independently perform data extraction. Authors of reviewed papers will be contacted in case of any missing details about their studies.

Data synthesis

A meta-analysis of outcomes combining various studies included in the review shall be done. We will assess statistical heterogeneity with I^2, which will indicate the percentage of the total variation

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across studies: 0% - 40% low heterogeneity, 30% - 60% moderate heterogeneity, 50% - 90% may represent substantial heterogeneity, and 75% - 100% is considerable heterogeneity. If there is a substantial amount of heterogeneity (75%), then sources of heterogeneity will be examined through subgroup and sensitivity analyses. We will also use Chi-square test to test the heterogeneity and consider P-values < 0.05 as statistically significant. A fixed-effects model will be selected for significant homogeneous studies; otherwise we will apply a random-effects model. All outcomes will be summarized using odds ratios (OR) and 95% CI. An OR<1 will represent a lower rate of outcome among the group of patients who were treated following the guidelines. Publication bias will be assessed by visual inspections of funnel plots and Egger's test.

We will also provide a narrative synthesis of the findings from the included studies. The narrative synthesis shall be structured by describing the studies according to the type of intervention used. This will include the three categories of interventions recommend in the PUPPG guideline [9]:

1. Prevention of pressure ulcers, including risk factors and risk assessment, skin and tissue assessment, preventive skin care and emerging therapies for prevention of pressure ulcers;

2. Interventions for prevention and treatment of pressure ulcers, such as nutrition in pressure ulcer prevention and treatment, repositioning and early mobilization, repositioning to prevent and treat pressure ulcers, support surface and medical device related pressure ulcers;

3. Treatment of pressure ulcers, e.g. assessment of pressure ulcers and monitoring of healing, pain assessment and treatment, wound care, assessment and treatment of infection and biofilms, wound dressings for treatment of pressure ulcers and surgery for pressure ulcers. Results will be presented in tables, figures and graphs followed by discussion.

Publication bias will be assessed in all analyses synthesizing 10 or more studies to ensure adequate power in the analysis[18]. For investigation of the effect of small studies and publication bias, data from included studies will be entered into a funnel plot asymmetry test if we have at least 10 studies in the meta–analysis. Egger's statistical test will be implemented using STATA/SE version 13 (Stata Corp, College Station, TX). The quality of supporting evidence will be assessed by the Grades of Recommendation, Assessment, Development and Evaluation (GRADE)[19].

Patient and Public Involvement

No patient involved.

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Ethics and Dissemination:

This review will only use published literature and will not recruit participants. Therefore, no formal ethical approval or consent is necessary. It is anticipated that this systematic review will provide a detailed summary of the evidence of the effectiveness of the PUPPG in preventing the occurrence of pressure ulcers among elderly patients in hospital. It is also expected that the study will provide recommendations on the best pressure ulcer preventive strategies applicable in health care settings. We aim to publish the research findings in a peer–reviewed scientific journal to promote knowledge transfer, as well as in various media, such as: conferences, seminars, congresses or symposia in a traditional manner.

LIST OF ABBREVIATIONS

CI: Confidence Intervals

EPUAP: European Pressure Ulcer Advisory Panel

GRADE: Grading of Recommendations Assessment, Development and Evaluation

JBI: Joanna Briggs Institute

JBI-MASTARI: Joanna Briggs Institute Meta-Analysis of Statistics Assessment and Review instrument

NPUAP: National Pressure Ulcer Advisory Panel

OR: odds ratios

PPPIA: Pan Pacific Pressure Injury Alliance

PRISMA-P: Preferred Reporting Items for Systematic Reviews and Meta-Analyses Protocols

PROSPERO: International Prospective Register of Systematic Reviews

RCT: randomized control trial

PUPPG: Pressure Ulcer Prevention Practice Guidelines

DECLARATIONS

Amendments

Any subsequent modifications made to this protocol will be documented and updated via PROSPERO and stated in the final review manuscript.

Consent for publication: Consent to publish is not needed as data that shall be used in this review shall be available in the public domain. Also, we will not use any details, images or videos related to individual participants in the review.

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Availability of data and materials: Not applicable

Competing interests: The authors declare that they have no competing interests

Funding: This work was supported by Hunan Provincial Key Laboratory of Nursing, grant number (2017TP1004), Hunan Provincial Science and Technology Department, China.

Author Contributions

AB, HM and WC contributed to the conception of the research question and writing of the protocol. HM, AB, WC, MM and DF contributed to the development of search strategies, eligibility criteria and methodology for data synthesis. HM, AB, WC, MM and DF contributed to drafting of the protocol and provided approval for the final version of this protocol. HM, AB and WC will work in duplicate to screen the titles and abstracts of all the materials obtained using the search strategy to exclude the articles that do not meet the eligibility criteria. HM, AB and WC will evaluate the potentially eligible studies with the full text and further exclude studies with documentation of the reason for exclusion. All authors will contribute to the bias assessment strategy and data extraction criteria. HM, AB and WC will independently extract data from the included studies. HM, AB and WC will analyze the data and draft the manuscript. All authors will read, provide feedback and approve the final manuscript.

Acknowledgements: The authors would like to thank Lindsey Sikora (librarian) for counseling in developing the searching strategies.

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Appendices

Appendix 1

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☑	☐	Page2/21, line 8
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☐	
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☑	☐	Page 4/21, line 47
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☑	☐	Page 2/21, lines 10-55
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☑	☐	Page 13/21, lines 3-21
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☑	☐	Page 12/21, lines 36-41
Support					
Sources	5a	Indicate sources of financial or other support for the review	☑	☐	Page 12/21, lines 51-53
Sponsor	5b	Provide name for the review funder and/or sponsor	☑	☐	Page 12/21, lines 53
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☑	☐	Page 12/21, lines 51-53
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	☑	☐	Page 6-7/21
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☑	☐	Page 8/21, lines 3-14
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report	☑	☐	Page 8/21, lines 20-55;

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
		characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review			Page 9/21, lines 3-18
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☑	☐	Page 9/21, lines 32-40
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☑	☐	Page 9/21, lines 20-31
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☑	☐	Page 9/21, line 52
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☑	☐	Page 9/21, lines 48-55; page 10/21, lines 3-14
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☑	☐	Page 10/21, lines 26-50
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☑	☐	Page 8/21, lines 31-50
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☑	☐	Page 8/21, lines 52-55; page 9/21, lines 3-7
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☑	☐	Page 10/21, lines- 15-24
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☑	☐	Page 11/21, lines 3-43
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☑	☐	Page 11/21, lines 3-14

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☑	☐	Page 11-21,lines 14-21
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☑	☐	Page 11/21, lines 22-43
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	☑	☐	Page 12/21, lines 32-41
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	☑	☐	Page 12/21, lines 43-47

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Appendix 2

Searching Strategies and Results in Medline (Ovid)

Database: Ovid MEDLINE(R) and Epub Ahead of Print, In Process & Other Non-Indexed Citations and Daily <1946 to March 12, 2019>
Search Strategy:

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1 Pressure Ulcer/ (11656)
2 (pressure adj3 sore*).ti,ab. (3032)
3 (pressure adj3 ulcer*).ti,ab. (7268)
4 (bed adj3 sore*).ti,ab. (221)
5 (decubitus adj3 ulcer*).ti,ab. (1787)
6 (pressure adj3 injur*).ti,ab. (2048)
7 or/1-6 (17175)
8 aged/ or "aged, 80 and over"/ (2913834)
9 elder*.ti,ab. (237584)
10 aged.ti,ab. (516384)
11 geriatric*.ti,ab. (43081)
12 or/8-11 (3320603)
13 Practice Guideline/ or Guideline/ or Guideline Adherence/ (60850)
14 guideline*.ti,ab. (306303)
15 pathway*.ti,ab. (997675)
16 protocol*.ti,ab. (390728)
17 PUPPG.ti,ab. (0)
18 recommendation*.ti,ab. (227666)
19 or/13-18 (1857233)
20 7 and 12 and 19 (544)

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Appendix 3

Joanna Briggs Institute for Meta-Analysis of Statistics Assessment and Review Instruments

1. JBI critical appraisal checklist for randomized control trials

Reviewer…….. Date……..

Author…….. Year……..

		Yes	No	Unclear	Not applicable
1	Was the assignment and treatment groups truly random?				
2	Were participants blind to treatment allocation?				
3	Was the allocation to treatment groups concealed from the allocator?				
4	Were the outcomes of people who withdrew described and included in the analysis?				
5	Were those assessing outcomes blind to the treatment allocation?				
6	Were the control and treatment groups comparable at entry?				
7	Were groups treated identically other than for the named interventions?				
8	Were outcomes measured in the same way for all groups?				
9	Were outcomes measured in a reliable way?				
10	Was appropriate statistical analysis used?				

Overall appraisal: include…… exclude……. Seek further info…….

Comments (include reasons for exclusion)……..

2. JBI critical appraisal checklist for non-randomized control trials

Reviewer…….. Date……..

Author…….. Year……..

1	Was study based on random or pseudo sample?				

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2	Were the criteria for inclusion in the sample clearly defined?				
3	Were confounding factors identified and strategies to deal with them stated?				
4	Were outcomes assessed using objective criteria?				
5	If comparisons are being made, was there sufficient description of the groups?				
6	Was follow up carried out over a sufficient time period?				
7	Were the outcomes of people who withdrew described and included in the analysis?				
8	Were outcomes measured in a reliable way?				
9	Was appropriate statistical analysis used?				

Overall appraisal: include…… exclude……. Seek further info…….

Comments (include reasons for exclusion)……..

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Appendix 4

Data Extraction Form for Quantitative Research

Study Characteristics

1. **Author:**
2. **Year of publication:**
3. **Journal:**
4. **Record number:**

Study Method

1. **Study Design:**

RCT	Quasi-RCT	Longitudinal	Retrospective	Observational	Other

2. **Purpose:**
3. **Research Questions:**

Participant Characteristics

Country where the study took place	Setting	Population	Sex	Age	Sample size	Ethnicity (if applicable)	socioeconomic status (if applicable)	education level (if applicable)

Pressure Ulcer Prevention Strategies

Strategy 1:

Strategy 2:

Strategy 3:

Outcome Measures

Outcome Description	Scale/measure

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Study results

Dichotomous data

Outcome	Intervention ( ) number / total number	Intervention ( )number / total number

Continuous data

Outcome	Intervention ( ) mean & SD (number)	Intervention ( ) mean & SD (number)

Authors' conclusions:

Comments:

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PRISMA-P 2015 Checklist

This checklist has been adapted for use with protocol submissions to *Systematic Reviews* from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☑	☐	Page2/21, line 8
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☐	
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☑	☐	Page 4/21, line 47
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☑	☐	Page 2/21, lines 10-55
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☑	☐	Page 13/21, lines 3-21
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☑	☐	Page 12/21, lines 36-41
Support					
Sources	5a	Indicate sources of financial or other support for the review	☑	☐	Page 12/21, lines 51-53
Sponsor	5b	Provide name for the review funder and/or sponsor	☑	☐	Page 12/21, lines 53
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☑	☐	Page 12/21, lines 51-53
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	☑	☐	Page 6-7/21
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☑	☐	Page 8/21, lines 3-14

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Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☑	☐	Page 8/21, lines 20-55; Page 9/21, lines 3-18
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